

TEJAS PATTABHI

Staff Engineer — Cloud · Distributed Systems · Applied AI

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SUMMARY

Staff Engineer with 10+ years turning large-scale data and compute problems into reliable production systems. At Qualcomm, works on the critical path of the autonomous-driving program — the infrastructure and applied AI that validate the software running in production vehicles. Named co-inventor on two granted U.S. patents.

CORE SKILLS

Languages	Python · Java · C# · C · Bash	Cloud	AWS · Google Cloud (Dataflow) · Azure
Orchestration	Kubernetes · Argo Workflows/Events · Docker · AWS CDK	Applied AI	LLM agents (Claude Agent SDK · Bedrock) · agentic coding (Claude Code · Codex) · custom Claude skills/plugins
ML & vectors	Self-hosted embeddings (Qwen3) · pgvector · semantic dedup		

EXPERIENCE

Qualcomm

Sep 2021 – Present

Staff Engineer · San Diego, CA

Build the large-scale infrastructure that replays recorded drive data to validate automated-driving software — perception, radar, vehicle self-motion/positioning, ground-truth labeling, and automatic emergency braking.

- **Automotive-grade reprocessing clusters** — designed, built, and operate two on-prem Kubernetes clusters of Qualcomm **QCR-100** cards (the same Snapdragon Ride silicon that ships in vehicles) for bit-accurate ADAS/AD reprocessing, running **~100–300x faster than general-purpose compute** — fast enough to replay recorded drives at close to real time.
- **Reprocessing-as-a-Service platform** running identically in cloud or on-prem — Kubernetes job orchestration, DAG dependency resolution, event-driven **Argo Workflows**, parallel petabyte-scale data transfer, and automated failure classification. Also integrated the QCR-100 into custom hosted cloud environments for on-demand partner/customer access.
- **Key collaborator on the Qualcomm–BMW Group program** — helped deliver the reprocessing pipeline behind the jointly developed Snapdragon Ride Pilot system (publicly launched Sept. 2025, debuting in the BMW iX3).
- **Production AI triage agent** (project lead) handling tens of thousands of failures/day — self-hosted embeddings + **pgvector** semantic dedup, then a **Claude Agent SDK + AWS Bedrock** agent that diagnoses root cause and opens Jira tickets / GitHub fix-PRs, escalating ambiguous cases to humans.
- **Fleet-health & accuracy monitoring** — Prometheus stack tracking each accelerator's health/firmware and **Real-Time Factor (RTF)**, the measure of how closely reprocessing matches ground truth; plus IPMI/BMC power control and safe Kubernetes node lifecycle for servicing without disrupting jobs.

OpenText (formerly Webroot)

Sep 2015 – Sep 2021

Senior Software Engineer · San Diego, CA

Led internet-scale cybersecurity and threat-intelligence systems.

- Led distributed clusters that safely crawled **17M+ URLs/day**; co-invented "DeepCrawl," which discovers related malicious resources from a known link and classifies threat level with an ML model — **basis of two granted U.S. patents**.
- Built an IP threat-classification system and a cloud-reputation scoring system, and architected multi-cloud honeypots (AWS, GCP, Azure) capturing live attacks to feed both. Owned AWS hosting/DevOps.

Earlier: Award Solutions — UT Dallas OpenStack project, open-sourced (2015) · TradeStation Technologies — SDE Intern, AWS EMR log analytics (2014) · Société Générale — Windows Server Administrator, Bangalore (2012–2013).

PATENTS · PUBLICATIONS · EDUCATION

U.S. Patents 12,452,259 B2 (2025) & 11,201,875 B2 (2021) — "Web Threat Investigation Using Advanced Web Crawling," named co-inventor, Open Text Inc. · First-author publication, *IJITCS*, 2013 (spatial-clustering algorithm).

M.S. Computer Science — UT Dallas · 2015

B.E. Information Science — PES / VTU, India · 2012 · First Class with Distinction (all 8 semesters)